

Jason Wang

jason.wang.25@ucl.ac.uk | linkedin.com/in/-jasonwang/ | github.com/wjason00

Computational developmental neuroscience — mechanistic modelling, scientific software, and reproducible analysis.

Education

University College London <i>Bachelor of Science in Neuroscience</i>	London Sept 2025 – Present
Calday Grange Grammar School <i>Mathematics (A*), Biology (A*), Chemistry (A), Further Mathematics (A)</i>	Wirral Sept 2023 – Jun 2025

Experience & Leadership

Private Tutor On-site bespoke tutoring	Wirral Aug 2022 – Aug 2025
• Provided three years of individual tutoring for Years 8–11 across Biology, Chemistry, Mathematics, Further Mathematics, Physics, Mandarin and Design & Technology. • Designed lessons around each student's starting point and led extended group sessions of 5+ hours, adapting explanations and questions to sustain engagement across ability levels.	

Projects

MorphBrain TypeScript, React, three.js / R3F, Zustand	2026 – Present
• Built an in-browser developmental-neuroscience demonstrator that solves deterministic SHH/BMP reaction–diffusion fields and maps them through a project-authored positional decoder to broad dorsal, intermediate and ventral tissue identities. • Separated mechanistic output from CC0 population-average dHCP surface-atlas playback (gestational weeks 21–36), exposing assumptions, provenance and limits in the interface; verified 549 automated tests across 86 files under strict TypeScript, lint and production-build gates.	
Model Morpher Python, PyTorch/CUDA, BrainGlobe GitHub	Dec 2025
• Built a GPU-accelerated tool for smooth morphological transitions between 3D brain structures, integrating the BrainGlobe Atlas API for interactive selection of regions from mouse-brain atlases (Allen / MPIN / CFL). • Implemented a batched PyTorch/CUDA pipeline computing signed distance fields on 128^3 grids, with linear SDF interpolation and marching-cubes isosurface extraction per frame. • Applied volume-preserving Taubin smoothing and mesh decimation to $\sim 15,000$-face outputs; real-time wire-frame preview in a configurable PyQt5 GUI.	
Morphliner TypeScript, WebGPU/WGSL, React, three.js morphliner.com	Jan 2026 – Present
• Developing a browser-native 3D mesh-morphing product whose WebGPU compute shaders perform signed-distance-field morphing and marching-cubes extraction, with a CPU fallback for unsupported hardware. • Extended the Model Morpher pipeline with timeline editing, STL/OBJ/PLY/GLB/GLTF import, local project persistence and in-browser MP4/WebM/GIF/PNG export at up to 4K.	

Achievements

British Biology Olympiad — Gold: top 5% of $\sim 18,500$ students.
Intermediate Biology Olympiad — Gold: only gold in school; top 5% of $\sim 15,000$ students.
Oxford University Computing Challenge: top 30 nationally (out of $\sim 20,000$ students).
Reply Challenge: top 0.5% globally in a 4-hour team coding challenge, collaboratively solving algorithmic tasks.

Technical Skills

Languages: Python, TypeScript, WGSL
Numerical & ML: finite-difference methods, reaction–diffusion systems, PyTorch, CUDA, NumPy, Pandas, scikit-learn
Graphics & 3D: marching cubes & signed-distance-function methods, PyVista, three.js / React Three Fiber, OpenCV, BrainGlobe Atlas API
Software: Git / GitHub, React, Vite, Zustand, Vitest, WebGPU, FastAPI, PyQt5, Fusion 360, Ultimaker CURA